Impacts of Simulated Coastal Ocean Alkalinity Enhancement on the Seasonal Cycle of Air-Sea CO_2 Exchange and Surface Ocean pCO_2 in European Waters under Low- and High-Emission Scenarios

Chiara Ciscato [1,3], David P. Keller[1,2], Neha Mehendale[1,4], Tronje Kemena[1]

[1]GEOMAR Helmholtz Centre for Ocean Research, Kiel (Germany)
[2]Carbon to Sea Initiative, Washington, DC (USA)
[3]CMCC Foundation - Euro-Mediterranean Center on Climate Change (Bologna)
[4]Institute for Environmental Physics, University of Bremen (Germany)

Key Points:

- Ocean Alkalinity Enhancement (OAE) is a carbon dioxide removal (CDR) strategy that aims to increase the ocean potential to sequester atmospheric CO_2
- OAE dampens the surface ocean pCO_2 seasonal cycle, especially under low warming
- OAE amplifies the seasonal cycle of the CO_2 flux, especially under high warming
- Regions with shallow, well-mixed waters may be ideal to implement OAE

current word count : ca 4,217 + 4 figures (each accounts for 500 words), limit for GRL is 6,000 words

Corresponding author: Chiara Ciscato, `chiara.ciscato@cmcc.it`

Abstract

To meet the Paris Agreement, emission reduction must be complemented by carbon dioxide (CO_2) removal technologies aiming to sequester atmospheric CO_2 and store it in permanent natural reservoirs. One potentially scalable method is ocean alkalinity enhancement (OAE), which works to lower ocean pCO_2 and accelerate CO_2 sequestration and storage. This study explores how OAE might affect the seasonal carbon cycle, which plays a key role in the ocean's annual CO_2 uptake. By analysing earth system model (ESM) simulations of OAE implemented continuously at the European coastline with a high and a low climate change forcing, it was found that an alkalised ocean: a) has a dampening effect on the seasonal cycle of the surface ocean pCO_2 as its buffering capacity is increased and CO_2 flux sensitivity reduced; b) has an amplification effect on the seasonal cycle of the CO_2 flux as it induces largest CO_2 uptake increase in the season when it is naturally highest; c) while the reduction of the ocean pCO_2 seasonal cycle is stronger under low warming, the CO_2 flux seasonal amplification is stronger in the high warming pathway.

Plain Language Summary

Although emission reduction remains the main strategy to mitigate human-induced global warming, technologies are being developed to sequester excess carbon dioxide (CO_2) from the air. Among the proposed CO_2 removal options is ocean alkalinity enhancement (OAE), which involves the addition of alkaline minerals or solutions at the ocean surface to lower the CO_2 concentration in seawater and accelerate its uptake from the atmosphere as a compensating effect. As seasonality is important to understand the annual carbon ~~movement~~, this research helps define the impacts of OAE implementation on the seasonal CO_2 cycle. To do this, a numerical earth system model (ESM) simulated continuous alkalinity addition at the European coastline, where it was found that adding alkalinity to the ocean's surface reduces its seasonal sensitivity and contributes to take up additional CO_2 from the atmosphere, especially over the winter months.

1 Research context

With its physical, chemical, and biological properties, the ocean constitutes a valuable resource for climate stabilisation, absorbing much of the earth's excess heat and storing vast amounts of carbon in its interior and deep waters (Friedlingstein et al., 2022; Scott-Buechler & Greene, 2019). With such premises, research on marine carbon dioxide removal (CDR) has been expanding, focusing on the use of wetlands, ~~bank~~ shelves and the open ocean to enhance natural CO_2 removal processes. One of the marine CDR under investigation is ocean alkalinity enhancement (OAE), which aims to speed up the ocean CO_2 uptake through accelerated chemical weathering, namely the breakdown of minerals by water-induced chemical reactions.

Theoretically, OAE would be performed by releasing alkaline minerals or solutions at the ocean surface, which would chemically convert aqueous CO_2 (CO_2(aq)) into stable bicarbonate (HCO_3^-) and carbonate (CO_3^{2-}) ions. This process would lower the ocean CO_2 partial pressure (pCO_2) and allow for additional CO_2 uptake from the atmosphere to restore the altered equilibrium. ~~While~~ increasing the ocean buffering capacity, OAE would have the additional benefit of reducing seawater acidity ~~by increasing its buffering capacity~~.

Although OAE has been widely investigated to account for theoretical benefits and challenges, little attention has been given to its effects on the seasonality of the carbon cycle, which is a major component of the annual net ocean uptake. OAE implementation could trigger significant seasonal impacts. For example, changes to the CO_2 flux downward might not be counterbalanced by compensating effects in the opposite direction, creating asymmetries throughout the year (Fassbender et al., 2022). Additionally, as CO_2 flux seasonality is a central driver of ocean extreme events such as earlier calcium carbonate undersaturation (Kwiatkowski et al., 2022) and intensified ocean acidification (Sasse et al., 2015), other than acting upon inter-annual variability patterns (Rustogi et al., 2023), unknown looping mechanisms may affect marine ecosystems at different scales. Lastly, while atmospheric pCO_2 responds to system stimuli at much shorter timescales, ~~the ocean~~ can take several seasons to react (Salt et al., 2013) and understanding such time lags is necessary for future OAE application.

This study explores carbon dynamics using output data from an Earth System Model (ESM) that simulated OAE along the European coastline under a low-emission and a high-emission socio-economic pathway (SSP) (SSP1-2.6 and SSP3-7.0, respectively). Our analysis aims to (a) define the impacts of OAE on the seasonal cycle of the CO_2 flux and surface ocean pCO_2, and (b) evaluate the influence of the background climate scenario on such outcomes.

2 Methods

2.1 Earth System Model

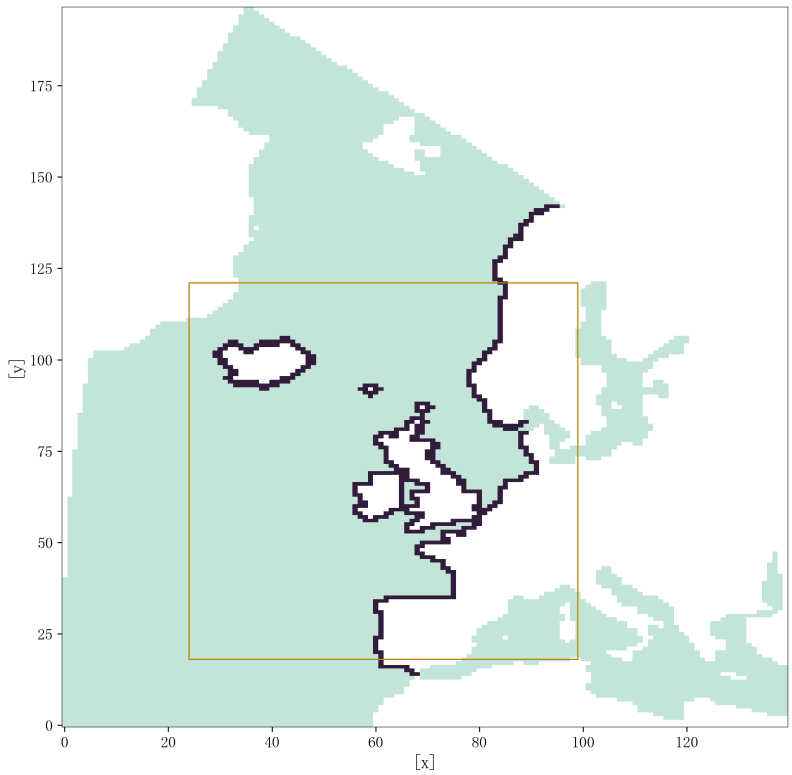
The data used in this manuscript was derived from the ESM Flexible Ocean and Climate Infrastructure (FOCI) (Matthes et al., 2020). The model consists of an atmosphere–ocean–sea ice general circulation model with a surface land component. The atmosphere is divided into 95 vertical layers and has a horizontal grid resolution of ∼1.8°, while the ocean is composed of 46 vertical layers and has a horizontal resolution of 1/2°.

FOCI incorporates the Model of Oceanic Pelagic Stoichiometry (MOPS) (Kriest & Oschlies, 2015) to enable the simulation of marine biogeochemical cycles, including a full carbon cycle. This allows to resolve biological dynamics such as phytoplankton uptake and remineralisation, calcite formation and dissolution (Chien et al., 2022).

2.2 Experimental design

Historical simulations were run from year 1850 to year 2015, when the SSPs begin (Riahi et al., 2017), and both the baseline and scenario forcings follow the CMIP6 protocol (O'Neill et al., 2016). In order to ~~define~~ the role of the future climate change scenario on OAE application, a low- and a high-emission SSP have been selected (SSP1-2.6 and SSP3-7.0, respectively) and simulations were conducted in emission-driven mode (O'Neill et al., 2016).

Figure 1: Geographical OAE implementation. The black bordering line represents alkalinity addition over a 50-km wide strip and the yellow rectangle defines the cropped area analysed in this manuscript.



In FOCI, both alkalinity and dissolved inorganic carbon (DIC) agree well with observations, while still showing a positive bias at the surface and a negative bias below 3000 meters of depth (Chien et al., 2022). Most DIC variations are due to organic matter build-up and CO_2 flux changes, while alkalinity is a prognostic tracer simulated as a combination of biological sources and sinks: nitrate and phosphate supply as sinks, calcium carbonate production and dissolution as sink and source, respectively, and organic matter (OM) formation and remineralisation, as sink and source, respectively (Chien et al., 2022).

In the OAE simulations, alkalinity addition is applied from year 2025 to year 2100. Alkalinity addition is subject to a linear increase over the first simulation decade (2025-2035), until the alkalinity equivalent of 1 Gt yr^{-1} of fast-reacting calcium hydroxide ($Ca(OH)_2$) is reached.

$$Ca(OH)_2 + 2CO_2 \leftrightarrow Ca^{2+} + 2HCO_3^- \tag{1}$$

Theoretically, $Ca(OH)_2$ consumes two moles of atmospheric CO_2 and produces one mole of calcium (Ca^{2+}) and two moles of bicarbonate, therefore increasing alkalinity by a factor of two (eq. (1)) (Chien et al., 2022). In our simulations, alkalinity is added to the system continuously and uniformly along a 50-kilometre-wide strip of the European coastline, with the exclusion of the Baltic and Mediterranean seas (fig. 1).

2.3 Data processing

The global output data delivered by FOCI was sliced to the European region, with coordinates 28°W to 12°E and 38°N to 70°N (yellow rectangle in fig. 1). Although the cropped area does not cover the whole region where OAE was performed, and part of the Norwegian and Spanish littoral were left out of the analysis, this approximation does not result in a meaningful modification of the results.

We present results for both the European region (yellow rectangle in fig. 1) and the addition coastline (black line in fig. 1). While the European region mean, that is, the mean over the yellow rectangle in fig. 1, is able to measure an approximated estimate of how the European system as a whole might respond to OAE implementation, the mean over the alkalinity addition coastline offers more precise insights into regime shifts that would take place at the epicentre of the domain.

Assuming that the most visible baseline-to-OAE differences are detected at the end of the century, the annual cycle averaged over the last simulation decade (2090-2099) is provided. While ocean pCO_2 and CO_2 flux are estimated only for the surface, alkalinity is measured over all 46 FOCI ocean layers. Thus, the variable was vertically averaged over a seasonally- and scenario-changing mixed layer depth (MLD) (also a FOCI output variable), which is the region considered to be in direct contact with the atmosphere and most favourable to CO_2 exchange. The alkalinity that is subducted below the MLD is assumed to not induce CO_2 uptake by the ocean, at least on shorter timescales.

3 Mechanisms driving the CO_2 flux seasonality

3.1 The thermal and the biophysical component

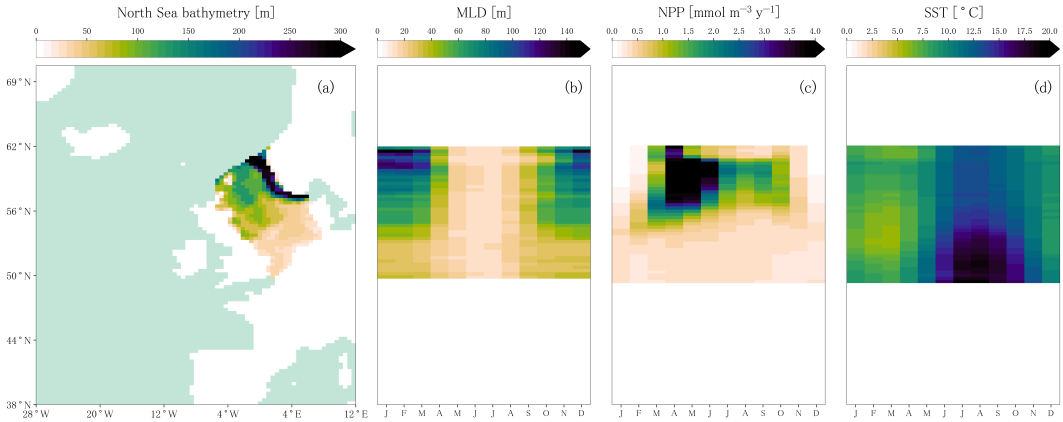
The flux of CO_2 across the air-sea interface is governed by the difference in the gas partial pressure between the atmosphere and the ocean: when atmospheric CO_2 exceeds oceanic CO_2, the flux is directed into the ocean, and viceversa. As OAE works to decrease the ocean CO_2 content, maximum efficiency is achieved when the deficit is restored by an equivalent amount of atmospheric CO_2 drawdown. If, however, the surface water residence time is shorter than the equilibration timescale, OAE efficiency is reduced.

Though with regional and local variations, the seasonal cycle of the CO_2 flux has two main components: sea surface temperature (SST), the thermal component, and DIC, the biophysical component (Fassbender et al., 2022). SST acts on gas solubility in seawater: over the winter months, cold temperatures favour CO_2 dissolution while warm temperatures in summer lead to CO_2 outgassing (Williams & Follows, 2011). As for DIC-driven seasonality, in winter, where processes like upwelling and OM respiration prevail and supersaturate the DIC pool, the ocean easily loses CO_2 to the atmosphere, whereas summertime OM production and downwelling favour atmospheric CO_2 uptake. As the dynamics induced by SST and DIC oppose each other (Lerner et al., 2021; Takahashi et al., 2002), the one prevailing drives the direction of the seasonal CO_2 exchange at a given location (Sarmiento & Gruber, 2006).

3.2 Seasonal drivers of the CO_2 flux in the study area

The air-sea CO_2 flux in the North Sea, where most of the OAE-induced changes occur (see Results section), is characterised by a strong North-to-South seasonal gradient (Salt et al., 2013; Prowe et al., 2009). Figure 2 show the zonally-averaged variables driving the seasonality of the CO_2 flux in the North Sea (highlighted region in fig. 2a), averaged over the last simulation decade in SSP1-2.6 baseline. Biological drivers dominate in the well stratified northern North Sea, where a relatively shallow mixed layer (fig. 2b) allows for nutrient-rich waters to stay at the surface and encourage phytoplankton bloom during the productive season (fig. 2c). OM that there accumulates then sinks below the surface layer, where remineralisation can happen far from atmospheric contact. At this location, CO_2 sequestration through carbon fixation outweighs temperature-driven outgassing, turning the area into a strong carbon sink (Salt et al., 2013; Prowe et al., 2009).

Figure 2: Map of the North Sea bathymetry (a) and hovmöller diagram of the zonally-averaged North Sea section for the SSP1-2.6 baseline seasonal cycle of the mixed layer depth (b), net primary production (NPP) (c), and sea surface temperature (SST) (d) over the 2090-2099 mean.



Conversely, in the southern North Sea and the adjacent European continental shelf, where a shallow water column (fig. 2a), and shallow MLD (fig. 2b) generate a more uniform layer all year round (fig. 2c) (Thomas et al., 2009; Hoppema, 1990), the air-sea CO_2 exchange is controlled by temperature variability (fig. 2d) (Salt et al., 2013). OM production and remineralisation happen within the same surface compartment, where CO_2 can easily be exchanged with the atmosphere (Prowe et al., 2009; Thomas et al., 2004). Here, where community respiration outweighs photosynthetic activity (Artioli et al., 2012), is a region of weak CO_2 outgassing (Prowe et al., 2009).

4 Results

In this section, results are provided for alkalinity, surface ocean pCO_2, and the CO_2 flux. The lineplots (fig. 3) represent the monthly cycle averaged over the last simulation decade in absolute values, together with the difference between the OAE scenario and the baseline (green line).

The maps (fig. 4) display the geographical variability of the seasonal amplitude change, defined as the difference between the annual maximum and the annual minimum. Note that calculations on the CO_2 flux are performed in the atmospheric component of FOCI, which has a coarser horizontal resolution.

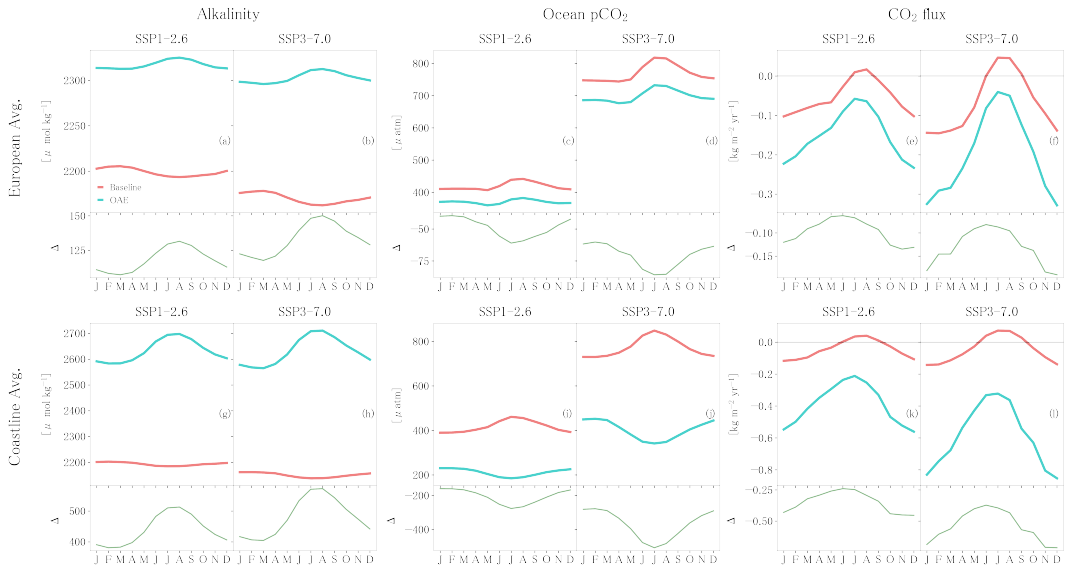
4.1 Surface and MLD-averaged alkalinity

For the European average in SSP1-2.6 (fig. 3a), baseline surface alkalinity registers lowest values in summer and highest in winter, with respective extremes corresponding to 2193 and 2205 μmol kg^{-1}. With alkalinity addition, absolute values rise at all seasons, with largest change happening in August. In SSP3-7.0, baseline values are lower than in SSP1-2.6, although OAE increases surface alkalinity relatively more than in the low emission scenario, with largest change happening in August as well.

At the coastline, OAE implementation forces a much higher alkalinity retention at the surface, with absolute values rising to a maximum of 2698 μmol kg^{-1} in SSP1-2.6 and 2711 μmol kg^{-1} in SSP3-7.0 in summer. Furthermore, the seasonal cycle is amplified greatly compared to the baseline (fig. 3g, h).

Under both warming scenarios and at both locations, alkalinity addition reverses its surface seasonal cycle, and summer values become much larger than winter values (fig. 3a, b, g, h). This reversing pattern is due to the fact that ocean stratification increases over warm months, which induces higher alkalinity retention at the surface where OAE is implemented.

Figure 3: Annual cycle averaged over 2090-2099 for surface alkalinity (a, b, g, and h), for ocean pCO_2 (c, d, i, and j), and for the CO_2 flux (e, f, k, and l), with all relative change (OAE-baseline). From left to right, columns indicate the European average (top) in SSP1-2.6 and in SSP3-7.0, and the coastline average (bottom) in SSP1-2.6 and in SSP3-7.0. In CO_2 flux plots, negative values indicate ocean uptake.



Under both SSPs, the alkalinity spatial pattern averaged over the MLD reveals strongest variation in the southern North Sea as well as by the UK coastline, where the seasonal amplitude increases greatly. SSP3-7.0 (fig. 4c) reveals an amplification signal compared to SSP1-2.6 (fig. 4a), while open ocean regions like Iceland, Spain and Norway show less strong modifications from the baseline.

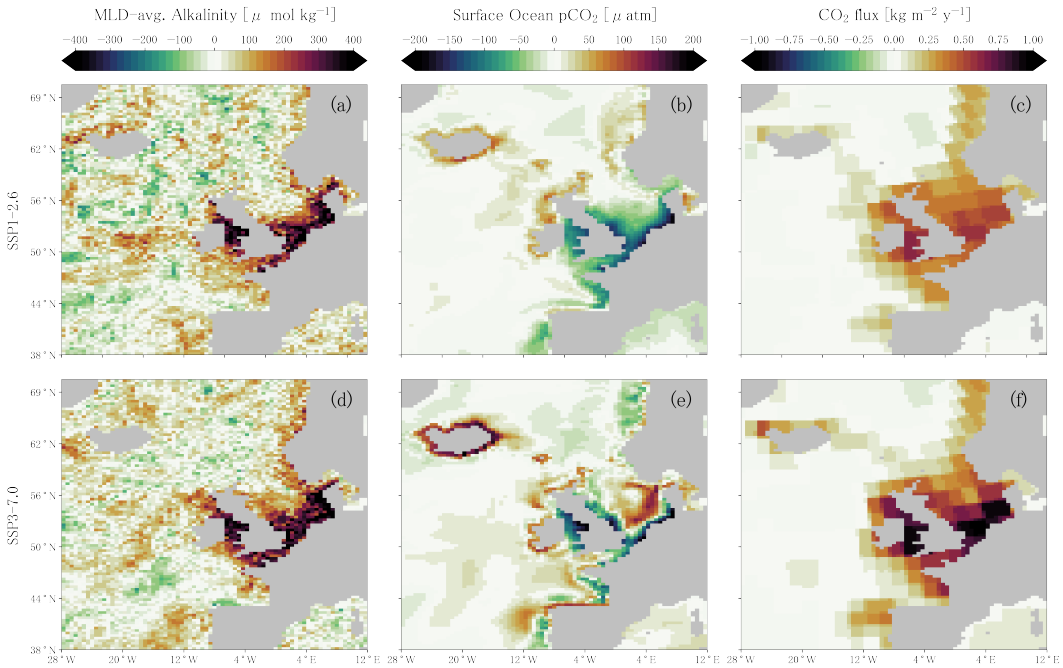
4.2 Surface ocean pCO_2

For the European average in SSP1-2.6 (fig. 3e), surface ocean pCO_2 is characterised by lowest and highest values in May and August, respectively, with a mean seasonal amplitude stretching from 406 µatm to 441 µatm. In the OAE ~~projection~~, ocean pCO_2 is reduced due to alkalinity enhancement and absolute values drop. The amplitude decreases, too, now spanning from 359 µatm to 382 µatm. Opposite to the lower warming scenario, where ocean pCO_2 is reduced due to lower atmospheric CO_2 concentration, in the baseline of SSP3-7.0 (fig. 3f), pCO_2 maintains an ascending trend until the end of the simulation (not shown). The mean seasonality has lowest and highest peaks of 744 µatm and 817 µatm, respectively, and OAE induces the seasonal amplitude to decrease substantially (55 µatm on average) (fig. 3).

At the coastline (fig. 3i, j), surface ocean pCO_2 is directly influenced by the reversal of the alkalinity seasonal cycle: highest (lowest) summer (winter) values in the baseline become lowest (highest) in the OAE scenario, while the seasonal amplitude is slightly reduced. In SSP1-2.6, surface ocean pCO_2 decreases by about 200 µatm, while in SSP3-7.0 the reduction is basically doubled and the OAE scenario reaches values similar to the SSP1-2.6 baseline, although the amplitude remains larger.

In both SSPs at both locations, OAE works to decrease the surface ocean pCO_2 seasonal cycle amplitude, as largest reduction takes place in summer, when ocean pCO_2 is highest. This outcome proves that alkalinity addition stimulates largest pCO_2 decline during the phase of CO_2 outgassing, at least in regions where CO_2 seasonality is temperature-driven like the European basin.

Figure 4: Seasonal amplitude change (baseline - OAE) averaged over 2090-2099 for alkalinity (a and d), for ocean pCO_2 (b and e) and for the CO_2 flux (c and f). The top row is scenario SSP1-2.6 and the bottom row is scenario SSP3-7.0.



In SSP1-2.6 (fig. 4b), a clear spatial pattern is identified, with two diverging geographies: at higher latitudes, including Iceland as well as the Norwegian and British North-West bank, the surface pCO_2 seasonal amplitude is slightly more pronounced than the baseline, while at lower latitudes, namely the mainland shoreline and the central-to-southern UK coasts, the seasonal cycle is strongly mitigated. In SSP3-7.0 (fig. 4e), the pCO_2 amplitude shows a similar behaviour, with the exception of the central North Sea and Northern Spain, where

strong seasonal amplification is displayed, and the Norwegian coast, where the ocean pCO_2 seasonal cycle is reduced.

4.3 CO_2 flux

As noted above, the CO_2 flux in fig. 3 shows that the European region CO_2 seasonality is overall driven by the thermal component, although with localised exceptions. Hence, the system tends to outgass (take up) CO_2 in summer (winter), at lowest (highest) chemical solubility. Largest CO_2 release happens between July and August (fig. 3e, f, k, l). Alkalinity addition enhances the CO_2 flux seasonal cycle everywhere and under both warming scenarios, with larger background emissions delivering greater amplification. In all Δ CO_2 graphs, the baseline-to-OAE differences grow in winter and decrease in summer, meaning that the impacts of alkalinisation on the ocean surface are largest when ocean uptake is already highest.

For the European average, OAE-induced CO_2 flux amplitude has an average span of 0.175 kg m^{-2} yr^{-1} in SSP1-2.6 and of 0.287 kg m^{-2} yr^{-1}, compared to their respective baselines (0.119 kg m^{-2} yr^{-1} in SSP1-2.6 and 0.192 kg m^{-2} yr^{-1} in SSP3-7.0).

In the baseline of both scenarios, at the coastline (fig. 3k, l), the annual cycle is split into two equally long phases. Outgassing takes place from May to October while CO_2 uptake from November to April. In the OAE-driven SSP1-2.6, CO_2 flux fluctuates between -0.5607 kg m^{-2} yr^{-1} and -0.2115 kg m^{-2} yr^{-1}, therefore more than doubling the baseline amplitude of 0.157 kg m^{-2} yr^{-1}. In SSP3-7.0, the seasonal amplitude of the OAE CO_2 flux is more than two times larger than the baseline, with a current amplitude of 0.531 kg m^{-2} yr^{-1}.

Spatially, in SSP1-2.6, the OAE scenario reveals the largest increase of the CO_2 seasonal flux along the European continental coastline, and the signal remains strong until the connection with the open ocean (fig. 4c), a pattern that is amplified under high emissions (fig. 4f), especially in the southern North Sea, in the UK and in North-West Iceland.

5 Discussion

5.1 Main drivers of OAE-induced seasonal changes

Figure 3 shows that seasonal system perturbations happen within the model domain, especially at the site that is in proximity of alkalinity addition, and the signal does not propagate far by the end of the century. This result is, however, not sufficient to assume that OAE is most successful near the addition site, as this manuscript does not address alkalised water that is subducted before air-sea equilibration is complete, and uncertainty remains on whether that water could resurface somewhere else to drive further CO_2 uptake (He & Tyka, 2023). In Palmiéri & Yool (2024), for example, it was estimated that about 50% of CO_2 uptake induced by global coastal OAE happened remotely from alkalinity injection sites.

The most affected areas are the North Sea, especially its southern part, and the coastal ocean between the UK and Ireland, which deliver pronounced seasonal alterations. This is probably due to the fact that a shallow, non-stratified and partially-enclosed ocean allows for alkalinity to accumulate at the surface, rather than being lost to deeper layers due to mixing-induced subduction (Liu et al., 2025; Palmiéri & Yool, 2024; Wang et al., 2023; Jones et al., 2014). Additionally, the seasonal cycle of surface alkalinity is reversed in all simulations, due to higher alkalinity accumulation at the top ocean layer over the summer months, when stratification is enhanced.

Ocean pCO_2 minima (and pH maxima, not shown) are registered in spring, as a result of phytoplankton bloom and consequent high photosynthetic activity (fig. 2c), where CO_2 is consumed (Gypens et al., 2011). In the OAE simulations, ocean pCO_2 seasonal amplitude is reduced as alkalised water minimises pCO_2 sensitivity to DIC changes, hence strengthening the ocean's buffering capacity. Regions like Iceland and Northern UK (and the central North Sea in SSP3-7.0) constitute an exception, as alkalinity addition causes an amplification signal of the pCO_2 seasonal cycle (fig. 4b,e). This may be due to the diverging OAE-induced reactions between a semi-enclosed space and the open ocean, where different physical processes occur.

Additionally, during OAE deployment, pCO_2 levels plummet and OAE-induced change is highest in the season where the associated seasonal driver, which is temperature for Europe, operates to reduce the ocean CO_2 sink potential.

In agreement with previous estimates (Schwinger, 2022), OAE works to amplify the CO_2 seasonal flux and take up more CO_2 regardless of the warming scenario (Nagwekar et al., 2024). OAE prompts a greater CO_2 flux downward in wintertime, meaning that OAE

induces an asymmetrical regime shift, with much lower lows than lower highs (where lower means higher ocean uptake). This is due to the fact that OAE effects are larger in the season where higher CO_2 drawdown from the atmosphere occurs naturally, hence in winter for Europe.

In both SSPs, highest alkalinity levels are recorded between the end of summer and the onset of autumn but largest uptake occurs in winter, therefore introducing a response lag of few months in re-equilibration timescales. Following (Jones et al., 2014), air-sea equilibration can take from a few months to even years to complete, depending on regional properties (Jones et al., 2014). Considering the features of the southern North Sea, where CO_2 flux modulations are dominated by temperature changes (Rodgers, 2023; Salt et al., 2013; Prowe et al., 2009) combined with a shallow water column, uniform mixing is likely to encourage fast equilibration with the atmosphere and to allow for efficient carbon sequestration under enhanced alkalinity.

5.2 Influence of the background climate state

When analysing the influence of the background climate state on OAE application, some key results can be highlighted. In OAE simulations, alkalinity accumulation at the top ocean layer happens under both SSPs, especially in summer. However, surface retention relative to the baseline is larger under higher warming, as increasing temperatures enhance ocean vertical stratification.

Though surface ocean pCO_2 absolute values show a less significant decline in SSP1-2.6, the seasonal compression signal is much stronger in this scenario, while in SSP3-7.0, pCO_2 rises (and pH falls) progressively until the end of the simulation, even with alkalinity enhancement (not shown). Conversely, the seasonal amplification of the CO_2 flux is amplified in SSP1-2.6. As noted by Nagwekar et al. (2024), this is probably more linked to the ocean carbonate chemistry, which is more sensitive under high CO_2 emissions, than to the background atmospheric CO_2 concentration.

Two key insights emerge from these results. First, the sensitivity of the ocean pCO_2 increases with more intense global warming and OAE alone is less efficient in mitigating the seasonal changes compared to a low-emission scenario. Second, in regions like the one analysed in this manuscript, rising temperatures, which have an amplifying effect on the thermally-driven component of ocean pCO_2 (Fassbender et al., 2022), lead to an overall expansion of the CO_2 seasonal cycle, exacerbated by OAE implementation.

6 Conclusions

This study addressed the OAE-induced changes on the seasonal carbon cycle in European waters, therefore contributing to the understanding of CO_2 intra-annual variability when performing continuous coastal ocean alkalinity enhancement. It was found that: (a) OAE dampens the ocean pCO_2 seasonal cycle and it amplifies the CO_2 flux cycle; (b) the former outcome is stronger in SSP1-2.6 while the latter turns stronger in SSP3-7.0; (c) locations like the southern North Sea could be ideal to implement OAE, as a shallow water column and intense mixing can fasten the CO_2 gas transfer at the air-sea interface, at least as far as the equilibration timescale is shorter than the surface water mean residence time.

Some limitations and recommendations for future research can be drawn from this analysis. Firstly, while freshwater input from rivers is modelled in FOCI, riverine alkalinity is not accounted for, limiting the reliability of resulting alkalinity absolute values. As river runoff features are deeply seasonal, implementing such component into future simulations would add complexity and precision to the system, ~~so to enhance the~~ understanding of ~~external sources to the~~ European coastal seas.

Secondly, our analysis does not separate ocean pCO_2 into its thermal and biological element, therefore missing to discern the influence that each of these have in altering the CO_2 seasonal cycle. Given the significance of temperature variability in the southern North Sea, taking each of the two terms individually could provide further insight into their seasonal role under OAE deployment.

Lastly, our OAE simulations pictures a highly idealised scenario, which is unlikely to become a real-world case. Other, more realistic OAE techniques should be explored, such as pulsed or point-source alkalinity addition. Building on these simulations, further investigation is recommended to build the premises for conscious large-scale OAE application.

7 Author contribution

Conceptualisation: C. Ciscato, D. P. Keller; Perfoming model simulations: T. Kemena; Data curation: C. Ciscato, D. P. Keller; Formal analysis: C. Ciscato; Investigation: C. Ciscato; Methodology: D. P. Keller, T. Kemena; Supervision: D. P. Keller; Validation: D. P. Keller; Visualisation: C. Ciscato; Writing – original draft: C. Ciscato; Writing – review & editing: C. Ciscato, D. P. Keller, N. Mehendale, T. Kemena

8 Acknowledgements

C.C. acknowledges funding from German BMBF, Project RETAKE. This is a contribution to the CDRmare research mission funded by the German Alliance for Marine Research (DAM). The authors wish to acknowledge use of Python for analysis and graphics in this paper.

9 Open Research

The code underlying this study is openly available at the following GitHub repository: 2024_ciscato_oae_seasonality

References

Artioli, Y., Blackford, J. C., Butenschön, M., Holt, J. T., Wakelin, S. L., Thomas, H., ... Allen, J. I. (2012). The carbonate system in the north sea: Sensitivity and model validation. *Journal of Marine Systems*, *102*, 1–13.

Chien, C.-T., Durgadoo, J. V., Ehlert, D., Frenger, I., Keller, D. P., Koeve, W., ... others (2022). Foci-mops v1–integration of marine biogeochemistry within the flexible ocean and climate infrastructure version 1 (foci 1) earth system model. *Geoscientific Model Development Discussions*, 1–58.

Fassbender, A. J., Schlunegger, S., Rodgers, K. B., & Dunne, J. P. (2022). Quantifying the role of seasonality in the marine carbon cycle feedback: an esm2m case study. *Global Biogeochemical Cycles*, *36*(6), e2021GB007018.

Friedlingstein, P., O'Sullivan, M., Jones, M. W., Andrew, R. M., Gregor, L., Hauck, J., ... others (2022). Global carbon budget 2022. *Earth System Science Data*, *14*(11), 4811–4900.

Gypens, N., Lacroix, G., Lancelot, C., & Borges, A. V. (2011). Seasonal and inter-annual variability of air–sea co2 fluxes and seawater carbonate chemistry in the southern north sea. *Progress in oceanography*, *88*(1-4), 59–77.

He, J., & Tyka, M. D. (2023). Limits and co 2 equilibration of near-coast alkalinity

enhancement. *Biogeosciences*, *20*(1), 27–43.

Hoppema, J. (1990). The distribution and seasonal variation of alkalinity in the southern bight of the north sea and in the western wadden sea. *Netherlands journal of sea research*, *26*(1), 11–23.

Jones, D. C., Ito, T., Takano, Y., & Hsu, W.-C. (2014). Spatial and seasonal variability of the air-sea equilibration timescale of carbon dioxide. *Global Biogeochemical Cycles*, *28*(11), 1163–1178.

Kriest, I., & Oschlies, A. (2015). *Mops-1.0: towards a model for the regulation of the global oceanic nitrogen budget by marine biogeochemical processes, geosci. model dev., 8, 2929–2957.*

Kwiatkowski, L., Torres, O., Aumont, O., & Orr, J. C. (2022). Modified future diurnal variability of the global surface ocean co2 system. *Global Change Biology*.

Lerner, P., Romanou, A., Kelley, M., Romanski, J., Ruedy, R., & Russell, G. (2021). Drivers of air-sea co2 flux seasonality and its long-term changes in the nasa-giss model cmip6 submission. *Journal of Advances in Modeling Earth Systems*, *13*(2), e2019MS002028.

Liu, F., Daewel, U., Kossack, J., Demir, K. T., Thomas, H., & Schrum, C. (2025). Evaluating ocean alkalinity enhancement as a carbon dioxide removal strategy in the north sea. *EGUsphere*, *2025*, 1–34.

Matthes, K., Biastoch, A., Wahl, S., Harlaß, J., Martin, T., Brücher, T., ... others (2020). The flexible ocean and climate infrastructure version 1 (foci1): mean state and variability. *Geoscientific Model Development*, *13*(6), 2533–2568.

Nagwekar, T., Nissen, C., & Hauck, J. (2024). Ocean alkalinity enhancement in deep water formation regions under low and high emission pathways. *Earth's Future*, *12*(10), e2023EF004213.

O'Neill, B. C., Tebaldi, C., Van Vuuren, D. P., Eyring, V., Friedlingstein, P., Hurtt, G., ... others (2016). The scenario model intercomparison project (scenariomip) for cmip6. *Geoscientific Model Development*, *9*(9), 3461–3482.

Palmiéri, J., & Yool, A. (2024). Global-scale evaluation of coastal ocean alkalinity enhancement in a fully coupled earth system model. *Earth's Future*, *12*(3), e2023EF004018.

Prowe, A. F., Thomas, H., Pätsch, J., Kühn, W., Bozec, Y., Schiettecatte, L.-S., ... de Baar, H. J. (2009). Mechanisms controlling the air–sea co2 flux in the north sea. *Continental Shelf Research*, *29*(15), 1801–1808.

Riahi, K., Van Vuuren, D. P., Kriegler, E., Edmonds, J., O'neill, B. C., Fujimori, S., ... others (2017). The shared socioeconomic pathways and their energy, land use, and greenhouse gas emissions implications: An overview. *Global environmental change*, *42*, 153–168.

Rodgers, K. (2023). *Seasonal variabiltiy of the surface ocean carbon cycle: a global synthesis* (Tech. Rep.). Copernicus Meetings.

Rustogi, P., Landschützer, P., Brune, S., & Baehr, J. (2023). The impact of seasonality on the annual air-sea carbon flux and its interannual variability. *npj Climate and Atmospheric Science*, *6*(1), 66.

Salt, L. A., Thomas, H., Prowe, A. F., Borges, A. V., Bozec, Y., & de Baar, H. J. (2013). Variability of north sea ph and co2 in response to north atlantic oscillation forcing. *Journal of Geophysical Research: Biogeosciences*, *118*(4), 1584–1592.

Sarmiento, J., & Gruber, N. (2006). *Ocean biogeochemical dynamics*. Princeton University Press.

Sasse, T., McNeil, B., Matear, R., & Lenton, A. (2015). Quantifying the influence of co 2 seasonality on future aragonite undersaturation onset. *Biogeosciences*, *12*(20), 6017–6031.

Schwinger, J. (2022). Report on modifications of ocean carbon cycle feedbacks under ocean alkalinization.

Scott-Buechler, C. M., & Greene, C. H. (2019). Role of the ocean in climate stabilization. *Bioenergy with Carbon Capture and Storage*, 109–130.

Takahashi, T., Sutherland, S. C., Sweeney, C., Poisson, A., Metzl, N., Tilbrook, B., ... others (2002). Global sea–air co2 flux based on climatological surface ocean pco2, and seasonal biological and temperature effects. *Deep Sea Research Part II: Topical Studies in Oceanography*, *49*(9-10), 1601–1622.

Thomas, H., Bozec, Y., Elkalay, K., & De Baar, H. J. (2004). Enhanced open ocean storage of co2 from shelf sea pumping. *Science*, *304*(5673), 1005–1008.

Thomas, H., Schiettecatte, L.-S., Suykens, K., Koné, Y., Shadwick, E., Prowe, A. F., ... Borges, A. (2009). Enhanced ocean carbon storage from anaerobic alkalinity generation in coastal sediments. *Biogeosciences*, *6*(2), 267–274.

Wang, H., Pilcher, D. J., Kearney, K. A., Cross, J. N., Shugart, O. M., Eisaman, M. D., & Carter, B. R. (2023). Simulated impact of ocean alkalinity enhancement on atmospheric co2 removal in the bering sea. *Earth's Future*, *11*(1), e2022EF002816.

Williams, R. G., & Follows, M. J. (2011). *Ocean dynamics and the carbon cycle: Principles and mechanisms*. Cambridge University Press.